

Cybersecurity Incident Report

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log

Based on my analysis of the DNS and ICMP traffic logs, the issue was caused by a failure in DNS communication between the client system and the DNS server.

The logs show that the client attempted to send DNS queries using the UDP protocol on port 53 in order to resolve the domain name `yummyrecipesforme.com`. However, instead of receiving a valid DNS response, the client received ICMP error messages indicating "UDP port 53 unreachable".

This error message signifies that the destination server was not accepting traffic on port 53, which is typically used for DNS services. This could be due to the DNS service being unavailable, misconfigured, or blocked by a firewall.

As a result of this failure, the domain name could not be resolved into an IP address, preventing the browser from establishing a connection to the web server. This ultimately led to the destination port unreachable error experienced by users.

In conclusion, the incident was caused by the inability to reach the DNS service over UDP port 53, as confirmed by the ICMP error responses, which disrupted the domain name resolution process.

Part 2: Explain your analysis of the data and provide at least one cause of the incident

From my analysis of the data, I observed that the client system was repeatedly sending DNS queries to the DNS server using UDP on port 53 to resolve the domain name `yummyrecipesforme.com`. These requests are visible in the `tcpdump` logs as outgoing packets from the client to the DNS server.

However, instead of receiving valid DNS responses, the client received ICMP error messages stating "UDP port 53 unreachable." This indicates that the DNS server was unable to process the request because the destination port required for DNS services was not accessible. The repeated ICMP responses confirm that each DNS request failed consistently.

This analysis shows that the DNS resolution process was interrupted, which prevented the system from obtaining the IP address of the website. As a result, the browser could not

proceed with establishing a connection to the web server.

One likely cause of this incident is that the DNS service on the server was either down, misconfigured, or not running, meaning no application was listening on port 53. Another possible cause could be that a firewall or network security control was blocking UDP traffic on port 53, preventing the requests from reaching the DNS service.

In conclusion, the data indicates that the failure of DNS communication over UDP port 53, confirmed by ICMP “port unreachable” messages, was the root cause of the incident.